Branch: MCA (AI/ML) Kargil	**Semester:** 2
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Experiment 9

Experiment Aim:

To implement database triggers in PostgreSQL to automatically calculate values and enforce constraints during data insertion operations.

Tools used: PostgreSQL

Objectives:

- **Trigger Implementation:** To understand how to create and use triggers in PostgreSQL.
- **Automation:** To automate calculation of total payable amount using triggers.
- **Data Integrity:** To enforce constraints using trigger conditions.
- **Real-time Processing:** To execute logic automatically before inserting data.

Theory:

Triggers are special database objects that automatically execute a function when a specified event (INSERT, UPDATE, DELETE) occurs on a table. In PostgreSQL, triggers are associated with trigger functions written in PL/pgSQL. They help maintain data integrity, enforce business rules, and automate tasks.

Experiment Steps:

Step 1: Creating Table **Query:**

```
CREATE TABLE employee (
emp_id INT PRIMARY KEY,
emp_name VARCHAR(50),
working_hours INT,
perhour_salary NUMERIC,
total_payable_amount NUMERIC
);
```

Step 2: Creating Trigger Function **Query:**

```
CREATE OR REPLACE FUNCTION CACULATE_PAYABLE_AMOUNT()
RETURNS TRIGGER
AS
$$
BEGIN
   NEW.total_payable_amount := NEW.perhour_salary * NEW.working_hours;
```

```
    IF NEW.total_payable_amount > 25000 THEN
        RAISE EXCEPTION 'INVALID ENTRY BEACUSE PAYABLE AMOUNT CAN NOT BE GREATER THAN 25000';
    END IF;

    RETURN NEW;
END;
$$ LANGUAGE PLPGSQL;
```

Step 3: Creating Trigger

Query:

```
CREATE OR REPLACE TRIGGER AUTOMATED_PAYABLE_AMOUNT_CALCULATION
BEFORE INSERT
ON employee
FOR EACH ROW
EXECUTE FUNCTION CACULATE_PAYABLE_AMOUNT();
```

Step 4: Inserting Valid Data

Query:

```
INSERT INTO EMPLOYEE(emp_id, emp_name, working_hours, perhour_salary)
VALUES (1, 'AKASH', 10, 1000);
```

Output:

```
Data Output   Messages   Notifications

INSERT 0 1

Query returned successfully in 1 secs 45 msec.
```

Step 5: Inserting Invalid Data (Exception Case)

Query:

INSERT INTO employee(emp_id, emp_name, working_hours, perhour_salary) VALUES (2, 'Ankush', 8, 100000);

Output:

```
Data Output   Messages   Notifications

ERROR:  INVALID ENTRY BEACUSE PAYABLE AMOUNT CAN NOT BE GREATER THAN 25000
CONTEXT:  PL/pgSQL function caculate_payable_amount() line 6 at RAISE

SQL state: P0001
```

Step 6: Retrieving Data

Query:

SELECT * FROM EMPLOYEE;

Output:

Data Output | Messages | Notifications

Showing rows: 1 to 1

	emp_id [PK] integer	emp_name character varying (50)	working_hours integer	perhour_salary numeric	total_payable_amount numeric
1	1	AKASH	10	1000	10000

Outcomes:

- Students will be able to create and use triggers in PostgreSQL.
- Students will understand how to automate calculations using triggers.
- Students will enforce constraints using trigger conditions.
- Students will understand real-time execution of database logic.